

Dark Data Forensics Vault

Complete Technical Languages & Systems Architecture Specification Report

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1. Core Backend Programming Language

Java (Java 21 LTS)

PRIMARY RUNTIME (100% OF BACKEND)

Component / File	Language Feature	Technical Role & Functional Implementation
ForensicDataVault.java	JCA MessageDigest Streams API	Ingests multi-modal files, parses Slack JSON & CSVs, calculates root SHA-256 digests, and generates recursive 1000-token chunks with 150-token overlaps.
ForensicQueryEngine.java	HttpClient (HTTP/2) LangChain4j	Local RAG orchestration with Ollama (mistral/llama3), Dynamic Pre-Flight Connection Guard (1200ms ping), and strict SHA-256 evidentiary prompt modifiers.
OnlineDataMapper.java	ByteArrayStreams Regex Parsers	In-memory RAM-only streaming client for U.S. SEC EDGAR filings (10-K / 8-K) with diskBypassed: true, leaving zero temporary disk footprints.
RiskAssessmentRecord.java	Java 21 Records	Immutable data carrier for detected corporate liabilities (e.g. \$18.5M Cayman SPE debt, CFO Slack concealment directives).
ForensicWebUI.java	Javalin 6.1 Jetty Microserver	REST API routing, dynamic port collision fallback (automatically resolves port 8080 collision and binds to port 8085).

2. Frontend & User Interface Technologies

Language	File Location	Implementation Purpose
HTML5	src/main/resources/public/index.html	Semantic layout for real-time telemetry gauges, the Operation BlackBriar fraud walkthrough, and query consoles.
CSS3	Embedded in index.html	Modern dark-mode aesthetic, glowing forensic status badges (Critical, High, Medium), animated loaders, and responsive grid layout.
JavaScript (ES6+)	Client script in index.html	Asynchronous fetch() calls to REST endpoints, real-time metrics polling, and interactive state management for the 4-step fraud case loader.

3. Data Serialization & Structured Interchange Formats

Format	Usage in Project	Functional Description
JSON	REST Payloads & Slack Dumps	Structured schema for REST endpoints (/api/stats, /api/query), Slack conversation dumps (internal_slack_dev_channel.json), and DevContainer configs.
CSV	Offshore Liability Ledgers	Tabular data extraction for unrecorded off-balance sheet transactions (offshore_unrecorded_liabilities.csv, fraud_case_offshore_spe_ledger.csv).
XML	pom.xml	Apache Maven Project Object Model defining dependencies, LangChain4j BOM, ONNX runtime, and build compilation plugins.

4. DevOps, Automation & System Scripting

⚙️ CI/CD & Deployment Languages

CLOUD AUTOMATION

Language	Workflow / Script	Execution Context
YAML	<code>.github/workflows/ci.yml</code>	GitHub Actions workflow automating JDK 21 installation, dependency caching, ONNX vector verification tests, and release JAR packaging.
PowerShell / Bash	<code>update_path.ps1</code> & Codespaces hooks	Portable Git and GitHub CLI (gh) configuration, environment variable manipulation, and 1-click cloud container startup scripts.

5. Native Machine Inference & Documentation Languages

💬 Native Engine & Documentation

PERFORMANCE & ARCHITECTURE

Language	Component	Technical Application
C++ (via JNI)	ONNX Runtime Native Engine	Pre-compiled C++ binaries accessed through Java Native Interface (JNI) to run the <code>bge-small-en-v1.5</code> neural embedding model at native machine speeds.
Markdown (GFM)	<code>README.md</code> , <code>enterprise_showcase.md</code>	Comprehensive technical documentation, Shields.io live badges, and 5-section executive pitch presentation.
Mermaid	Architecture diagrams in README	Diagram-as-code specification rendering interactive multi-agent topology flowcharts directly inside GitHub.